

SOFTWARE ENGINEERING

ASSIGNMENT – 02

PROJECT 1: SELECTION AND REQUIREMENTS ENGINEERING

TASK 1: TEAM SELECTION

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TASK 2: SOFTWARE REQUIREMENT SPECIFICATION(SRS)

1. Preface

This document describes the Software Requirements Specification for AdaptIQ, an AI-driven personalized study assistant. AdaptIQ is developed to help students enhance their learning outcomes by providing tailored recommendations based on their quiz performance.

The system uses machine learning algorithms to evaluate quiz results and identify areas where students excel and areas that require improvement. Based on this analysis, the system suggests relevant quizzes, practice exercises, and study materials.

Furthermore, AdaptIQ fosters a collaborative learning environment by enabling students to share their notes, discover resources created by peers, and work together on shared documents. This document outlines both functional and non-functional requirements, system models, and expected behavior. AdaptIQ aims to create an adaptive, efficient, and collaborative learning experience for students.

2.Introduction

2.1 Purpose of the System

AdaptIQ is designed as an AI-powered learning assistant that evaluates student quiz performance and provides personalized study recommendations using machine learning techniques. It also aims to build a community-driven repository of study materials through notes sharing and collaboration.

Its purpose is to help students study more efficiently by highlighting weak areas, offering suitable quizzes and study materials to strengthen knowledge in those topics, and providing access to peer-validated learning resources.

2.2 Scope of the System

AdaptIQ provides a smart learning environment that adjusts study content according to each student's performance and facilitates peer-to-peer learning.

The system will:

- Allow students to take quizzes on different subjects and topics.
- Evaluate quiz results using a machine learning model.
- Determine areas of strength and weakness.
- Recommend personalized quizzes to improve weak topics.
- Enable students to track learning progress.
- Present insights and study recommendations.
- Allow students to upload and share their own notes with descriptions and tags for the course/topic.
- Provide a search and filter functionality to find notes by subject, course, or topic.

- Enable a rating and review system for shared notes to highlight the most useful resources.
- Include collaboration tools for real-time, shared document editing among students.

The system will not:

- Replace comprehensive learning management platforms such as Moodle or Blackboard.
- Monitor classroom activities in real-time.
- Substitute human teachers or tutors.

The main focus is to provide adaptive learning, personalized study guidance, and a collaborative resource-sharing platform through AI and community features.

3. Glossary

- ❖ AI: Artificial Intelligence that enables systems to perform tasks similar to human decision-making.
 - ❖ Machine Learning: Algorithms that learn patterns from data and make predictions or decisions.
 - ❖ Adaptive Learning: A method where learning content is adjusted based on student performance.
 - ❖ Recommendation System: A system that provides users with suitable learning resources or suggestions.
 - ❖ Quiz Performance: The score achieved by a student on a quiz.
 - ❖ Dataset: A structured collection of data used to train machine learning models.
 - ❖ Notes Sharing: The functionality that allows users to upload and share their study notes with the community.
 - ❖ Collaboration Tools: Features that enable multiple users to work on a single document simultaneously.
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4. User Requirements Definition

4.1 Functional Requirements

- Users can create a personal study profile.
- Users can select subjects and topics to study.
- Users can attempt quizzes.
- The system will evaluate answers and calculate quiz scores automatically.
- Performance will be analyzed using a machine learning model.
- Weak topics will be identified based on quiz results.
- The system will suggest quizzes to improve weak areas.
- Users will receive study suggestions based on their performance.
- Users can monitor their learning progress over time.
- Users can review previous quiz attempts and results.
- Users can upload and share their study notes with titles, descriptions, and subject/course tags.
- Users can search for shared notes using keywords, subjects, courses, or topics.
- Users can rate and write reviews for notes shared by other students.
- Users can collaborate with peers on a shared document in real-time.

4.2 Non-Functional Requirements

1. **Usability:** The system should have a clear, simple, and user-friendly interface.
2. **Performance:** Recommendations should be generated within five seconds after a quiz is submitted.
3. **Accuracy:** The machine learning model should provide consistent and reliable suggestions.
4. **Security:** User data must be stored securely and protected from unauthorized access.

5. **Compatibility:** The system should function on modern web browsers and devices.
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5. System Requirements Specification

5.1 Functional System Requirements

User Profile Creation

Requirement ID: F-01

Description: The system enables users to create a study profile.

Input: User name and selected subject.

Output: Confirmation that the profile is successfully created.

Error Handling: If required information is missing, the system will display a message prompting completion.

Quiz Attempt

Requirement ID: F-02

Description: Users can take quizzes based on selected topics.

Processing Steps:

1. The user selects a subject or topic.
2. The system displays quiz questions.
3. The user answers the questions.
4. The user submits the quiz.

Output: Quiz scores are calculated and stored.

Performance Analysis

Requirement ID: F-03

Description: The system analyzes student performance using a machine learning model.

Processing Steps:

1. Collect quiz scores.
2. Process performance data.
3. Identify strong and weak areas.

Output: A performance report is generated.

Personalized Recommendation

Requirement ID: F-04

Description: The system recommends quizzes and study materials based on performance.

Processing Steps:

1. Identify weak areas from performance analysis.
2. Process data using the machine learning model.
3. Generate tailored recommendations.

Output: Recommended quizzes and learning materials are presented to the user.

Progress Tracking

Requirement ID: F-05

Description: The system tracks and displays student progress over time.

Output: Visual charts and indicators showing performance improvements.

Notes Sharing

Requirement ID: F-06

Description: The system allows users to upload and share their study notes with the community.

Search & Filter

Requirement ID: F-07

Description: The system enables users to find shared notes efficiently.

Processing Steps:

1. The user enters keywords or selects filter criteria (subject, course, topic).
2. The system queries the database for matching notes.

Output: A list of notes matching the search/filter criteria is displayed.

Ratings & Reviews

Requirement ID: F-08

Description: The system allows users to rate and review shared notes.

Processing Steps:

1. The user accesses a shared note.
2. The user provides a star rating and writes a text review.
3. The system saves the rating and review and updates the note's average rating.

Output: The new rating and review are displayed on the note's page.

Collaboration Tools

Requirement ID: F-09

Description: The system provides a real-time shared document editor for collaborative study.

Processing Steps:

1. The user creates a new collaborative document or invites peers to an existing one.
2. Multiple users can edit the document text simultaneously.
3. The system synchronizes all changes in real-time.

Output: A live document that all collaborators can view and edit together.

5.2 Non-Functional System Requirements

Usability

Requirement ID: NF-01

- Easy navigation
- Clear and organized quiz interface
- Simple and understandable performance feedback
- Intuitive design for uploading and searching notes

Performance

Requirement ID: NF-02

- Quiz evaluation within three seconds
- Recommendation generation within five seconds
- Search results for notes returned within two seconds
- Real-time synchronization of collaborative documents with minimal latency

Accuracy

Requirement ID: NF-03

- Recommendations generated by the machine learning model should be reliable
- Model trained on quiz performance datasets

Security and Privacy

Requirement ID: NF-04

- User data must be protected against unauthorized access
- Only essential information should be stored
- Access controls for collaborative documents to prevent unauthorized editing

Scalability

Requirement ID: NF-05

- Adding new subjects and quizzes should be simple
- Machine learning model should support future updates
- The system should handle a growing number of shared notes and concurrent users for collaboration

6. System Models

6.1 Use Case 1: Attempt Quiz

Actors

★ **Primary Actor:** Student

Preconditions: Students must have a registered profile.

Basic Flow:

1. The student selects a subject or topic.
2. Quiz questions are displayed.
3. Students answer the questions.
4. Quiz is submitted.

5. The system evaluates answers and calculates scores.

Postconditions: Quiz results are stored and used for performance analysis.

6.2 Use Case 2: Analyze Performance

Actors

★ **Primary Actor:** System

Preconditions: Students must have completed at least one quiz.

Basic Flow

1. The system retrieves quiz results.
2. The system analyzes performance for each topic.
3. Weak and strong areas are identified.

Postconditions: Performance report is generated.

6.3 Use Case 3: Generate Study Recommendation

Actors

★ **Primary Actor:** System

Preconditions: Performance analysis is completed.

Basic Flow

1. The system identifies weak areas.
2. Data is processed by the machine learning model.
3. Personalized study recommendations are generated.
4. Recommendations are displayed to the student.

Postconditions: Students receive customized study guidance.

6.4 Use Case 4: Share a Note

Actors

Primary Actor: Student

Preconditions: Student must be logged into their profile.

Basic Flow:

1. Student navigates to the "Upload Note" section.
2. Student selects a file to upload.
3. Student enters a title, description, and relevant tags.
4. Student submits the form.

The system saves the note and its metadata.

Postconditions: The note is added to the shared repository and becomes searchable by others.

6.5 Use Case 5: Collaborate on a Document

Actors

Primary Actor: Student

Secondary Actor: Collaborating Students

Preconditions: Students must be logged in and have initiated or joined a collaboration session.

Basic Flow:

1. Students create a new shared document or open a shared link.
2. Students and collaborators begin typing.
3. The system displays all participants' cursors and changes in real-time.
4. All students can view and edit the document content simultaneously.

Postconditions: A collaboratively created document is saved for all participants.

TASK 3: REQUIREMENT ELICITATION AND ANALYSIS

1. Elicitation Technique

The requirements for AdaptIQ were collected using brainstorming and analysis of existing systems.

The team first held brainstorming sessions to identify potential features for an AI-based study assistant. Since the project involves machine learning, ideas such as quiz performance analysis, adaptive learning, and recommendation systems were explored. This led to the creation of AdaptIQ as a personalized AI learning companion.

Next, existing learning platforms like Quizlet, Khan Academy, Google Docs, and other online quiz systems were analyzed. While these platforms provide quizzes and study materials, they often lack deep personalization. By studying these systems, the team recognized an opportunity to not only deliver targeted study recommendations using machine learning but also to combine this with a robust peer-learning environment. The popularity of shared resources and collaborative tools in education inspired the inclusion of the notes sharing, rating, and collaboration features.

These techniques were effective because they identified real student challenges (personalized learning, finding quality resources, collaborative studying) and ensured that the system addresses practical learning needs rather than adding unnecessary complexity.

2. Stakeholder Identification

Primary stakeholders are students, especially those in colleges and universities who take quizzes, prepare for exams, and form study groups. Students often struggle with identifying weak topics, planning study sessions effectively, receiving

personalized guidance, and finding high-quality study materials beyond their own notes. Many spend time on material they already know while neglecting areas that need more attention, or they miss out on the insights of their peers.

AdaptIQ addresses these issues by analyzing quiz performance and providing targeted quizzes and study materials using machine learning, while also creating a platform for students to benefit from the collective knowledge and resources of their peers through sharing and collaboration.

Secondary stakeholders include teachers and instructors, who aim to improve student learning outcomes. While they may not directly use the system, the recommendations provided by AdaptIQ support their teaching by helping students study more effectively. The shared notes and collaborative features also foster a richer learning environment outside the classroom.

Course evaluators and instructors are also stakeholders, as they expect the project to demonstrate meaningful use of machine learning. By incorporating ML-based performance analysis and recommendation features alongside comprehensive community-driven tools, AdaptIQ meets and exceeds these academic requirements.

TASK 4: GRADIO/STREAMLIT DEMO

Streamlit Demo:

[DemoLink](#)

GOOGLE Collab Streamlit URL:

<https://colab.research.google.com/drive/1ZsroDkS7EhoWicfu3wV2FjiyJIMcNZNn#scrollTo=ljznnSTYT-dx>